Homework 4

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1 STATA - hourly

1. Number of treated cohorts = 539

	# ATTs	$\sum$ Weights
Positive weights	145467	1.2453
Negative weights	48547	-0.2453
Total	194014	1.0000

Table 1: TWFE Weight Decomposition for Treatment Variable

2. Under the common trends assumption, the TWFE coefficient beta, equal to -0.0430, estimates a weighted sum of 194014 ATTs. 145467 ATTs receive a positive weight, and 48547 receive a negative weight.

3. The results are clustered at the household level.

	(1) Energy Consumption (kWh)
Treatment	-0.0434***
	(0.000232)
Temperature (F)	0.00461***
	(0.0000121)
Precipitation (in)	-0.000644
	(0.00200)
Relative Humidity (%)	0.00230***
	(0.00000576)
Constant	0.539***
	(0.00111)
Observations	720000

Standard errors in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 2: Regression results for hourly energy consumption

4. The treatment effect is much stronger at the daily level.

	(1)
	energy
Treatment	-0.936***
	(0.00557)
Temperature	0.111***
	(0.000379)
Precipitation	0.0681
	(0.188)
Relative Humidity	0.0552***
	(0.000189)
Constant	12.88***
	(0.0341)
N	30000

Standard errors in parentheses
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 3: Regression results for daily energy consumption

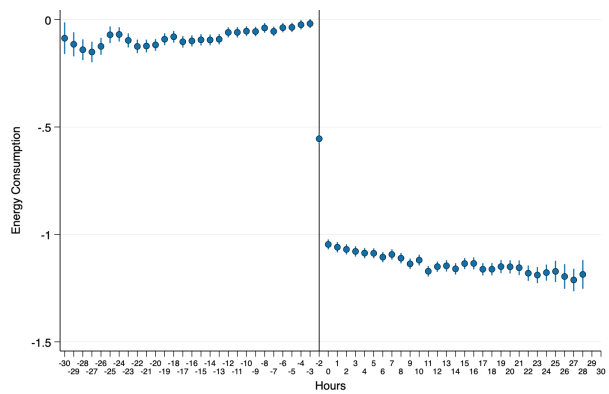

Figure 1: Event study 1 - Manually creating pre and post periods

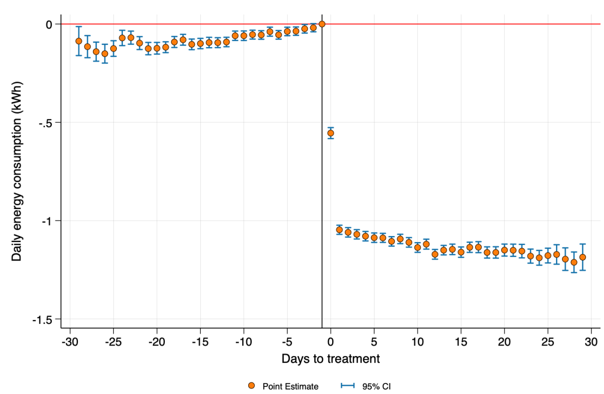

Figure 2: Event study 2 - Using eventdd

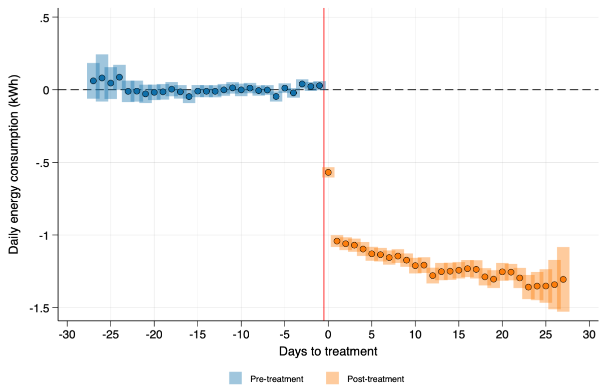

Figure 3: Event study 3 - Implementing Callaway and Sant'Anna estimator